

16. HOW ENVIRONMENTAL TEMPERATURE INFLUENCE ROAD CYCLING PERFORMANCE? MEN VS WOMEN DIFFERENCES?

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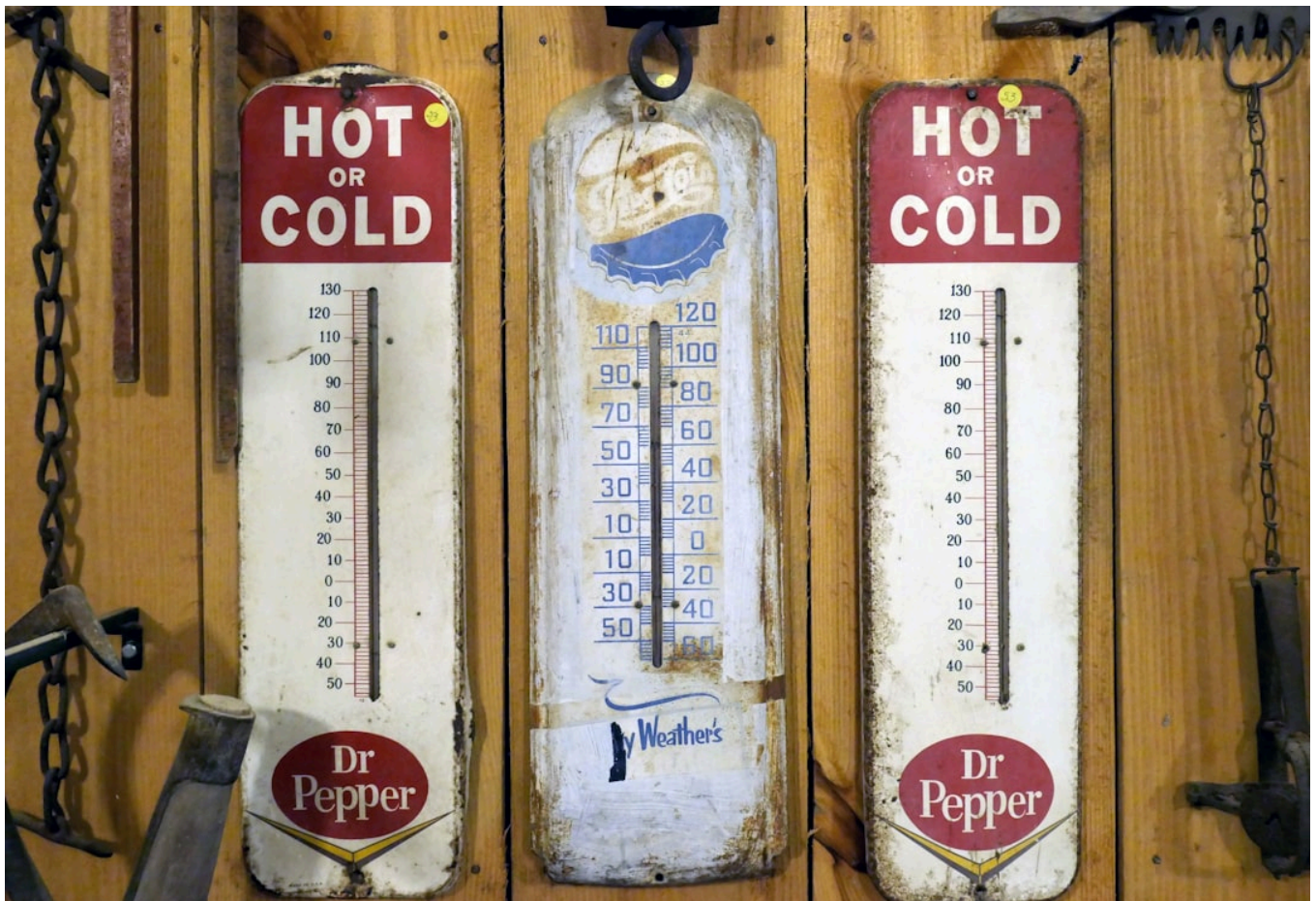


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A road cycling's unique features is to face all temperature conditions: from the extreme cold typical of some Giro d'Italia's alpine stages , to the scorching heat of Tour de France and Vuelta España. Laboratory studies showed as both low and high environmental temperatures negatively impact endurance performance. However, less is known about how much different temperatures influence professional road cyclists performance during training/competitions (**so on the road and not in the lab**).

Does pro road cyclist performance on the road decrease at very high or low temperatures? If yes, at which temperature and how much?

Is there any difference in men vs. women pro cyclists response to very high or low temperatures?

A study published by Valenzuela and Colleagues in 2022 on *International Journal of Sport Physiology and Performance* tried to answer these questions. (1)

WHAT DID THEY DO?

- They retrospectively analysed the maximal mean power outputs (MPOs) recorded during training and races over fixed duration (5s, 30s, 5min, 20min) of 48 males and 26 females pro road cyclists from 3 different teams. The timeframe analysed was between 2013 and 2020 season.
- Maximal Mean Power Outputs were divided for different environmental temperatures:
 - < 0°C
 - 0-5°C
 - 5-10°C
 - 10-15°C
 - 15-20°C
 - 20-25°C
 - 25-30°C
 - 30-35°C
 - >35°C
- Only riders with goal races both in winter and summer were included, to be sure they all performed maximal efforts at all temperature conditions.

**Maximal mean power outputs: MPOs. MPO5min means the maximum mean power output recorded over 5min.*

WHAT DID THEY FOUND?

- **Male Pro cyclists decreased MPOs below 10°C and above 30°C.**
- **Female Pro cyclists obtained their best MPOs at slightly lower temperatures:**
 - Short efforts (MPO5s and MPO30s) were impaired only below 5 °C (and not 10 °C like men)
 - Long efforts (MPO5min and MPO20min) decreased already above 25°C (and not only after 30 °C like men)

- **How much was the performance decrease at high and low temperatures?**
 - **2-7%** between 5-10°C or 30-35°C
 - **9-18%** at <5°C or >35°C

PRACTICAL TAKEAWAYS

- **The optimal range for cycling performance seems to be between about 10-30°C.**
- The further from this range, the greater the decrease in performance.
- Females seems to tolerate slightly better low temperatures, while worse high temperature, when compared to male cyclists.

GABRIELE'S COMMENT

How can we use these results?

- **When analysing a given performance**, keep in mind that a **decrease in power output is likely to happen outside the optimal ranges**. This could be also important to choose pacing strategy (for example the target power output to maintain during a TT).
- When planning training or analysing training sessions, consider the differences in temperature between the ambient in which power zones have been calculated and the ambient in which training is performed.
- Develop strategies to counteract the negative effect of low (warm up, equipment, ...) and high temperature (pre-cooling, per-cooling). Women could benefit from pre/per-cooling strategies already between 25-30°C.

Why female tolerate better lower temperature and worse high temperature? The lower sweating rates and the higher fat mass % observed in other studies could be the main factors.

In the next post we will have a look if also power and/or heart rate at thresholds and training zones change or not in the heat.

Thanks for Reading! Spread with your Team!

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REFERENCES:

1. Valenzuela PL, Mateo-March M, Zabala M, Muriel X, Lucia A, Barranco-Gil D, Pallarés JG. Ambient Temperature and Field-Based Cycling Performance: Insights From Male and Female Professional Cyclists. Int J Sports Physiol Perform. 2022 Mar 28;17(7):1025-1029.



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